REVA UNIVERSITY
Bengaluru, India

LLM-Powered Agentic Dashboard for Business-Intelligence

Name of the Presenter
SRN: R22MSA55 | Date: 27/06/2026

M.Sc. in
Artificial Intelligence
Capstone Project Presentation
Year: I

Agenda

General Instructions

NB: This slide is for reference only and must be removed while submitting the final deck. Text in Red in all slides is for reference only and must be removed.

1. Present a clear and logical flow – *Problem, Objectives, Methodology, Implementation, Results, and Conclusion.*
2. Keep slides visual, concise, and consistent in design; use diagrams, charts, and images instead of long text.
3. Highlight measurable outcomes and insights that directly link to project objectives.
4. Conclude with key learnings, future improvements, and acknowledgements.
5. Bring Design Thinking: You must bring design thinking principles while designing your deck.
    a. In the introductory section, bring the stakeholders' points of view and their pain points
    b. Highlight Human-Centred Approach: Show that your project started with understanding real user needs (Empathise → Define → Ideate → Prototype → Test).
    c. Link Stages to Project Flow: Briefly align each stage with what you did – problem understanding, idea generation, model prototyping, and user validation.
    d. Use Visual Storytelling: Add short visuals like personas, journey maps, and prototype screenshots to make your process clear and engaging.

Notes:

- Use Roboto Slab as a consistent font throughout
- You are free to change the standard template headings to customised ones as per your project.

Introduction

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Background

- Organizations generate large volumes of business data
- Traditional BI tools require manual dashboard design
- Analytics often depends on skilled analysts

Current Status

- LLMs can understand natural language queries
- AI-assisted analytics is rapidly evolving
- Most BI tools still lack seamless AI-driven dashboard generation

Why This Topic?

- Reduce time from question to insight
- Enable self-service analytics for non-technical users
- Automate visualization and insight generation using Agentic AI

Literature Review

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No.	Title	Author and Year	Journal	Description	Insights	Research Gaps/Comments
1	AI4VIS: Survey on Artificial Intelligence Approaches for Data Visualization	Wu et al. (2021)	IEEE TVCG	Comprehensive survey of AI techniques for data visualization, chart recommendation, and insight discovery.	Demonstrates the growing role of AI in visualization automation.	Does not provide an end-to-end natural language dashboard generation system.
2	MultiVision: Designing Analytical Dashboards with Deep Learning Based Recommendation	Wu et al. (2021)	IEEE TVCG	Recommends dashboard layouts using deep learning and visualization recommendation techniques.	Improves automated dashboard design.	Limited to dashboard recommendation; lacks analytics and workflow automation.

Literature Review

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3	Explainable Artificial Intelligence: A Survey of Needs, Techniques, Applications, and Future Direction	Mersha et al. (2024)	Neurocomputing	Reviews Explainable AI methods and interpretable AI systems.	Highlights the importance of generating human-understandable insights.	Does not integrate explainability with interactive BI dashboards.
4	nvAgent: Automated Data Visualization from Natural Language via Collaborative Agent Workflow	Ouyang et al. (2025)	ACL	Uses collaborative AI agents to convert natural language into data visualizations.	Shows the effectiveness of Agentic AI for visualization generation.	Focuses on visualization generation rather than a complete BI platform with preprocessing, validation, and dashboards.

Problem Statement

Businesses generate large amounts of data every day, but turning that data into meaningful insights still requires significant manual effort and technical expertise. Although AI can understand natural language, most business intelligence tools cannot automatically convert user questions into interactive dashboards with relevant visualizations and business insights, making data analysis slower and less accessible to non-technical users.

Key Challenges

- Manual data preparation and dashboard creation
- Dependence on technical expertise for data analysis
- Time-consuming analytical workflow
- Limited support for natural language-driven business intelligence

Project Objectives

1. **Design** an intelligent workflow for natural language-driven business data analysis and visualization.

2. **Develop** an interactive dashboard for dynamic data exploration and automated visualization generation.

3. **Evaluate** the accuracy, reliability, and effectiveness of generated visualizations and business insights.

4. **Deploy** an accessible business intelligence solution for intuitive decision support and reporting.

Project Methodology

Methodology: Team Data Science Process (TDSP)

Workflow:

- Business Understanding
- Data Acquisition & Understanding
- Data Preparation
- LLM Agentic Workflow
- Deployment
- Validation

Research Design

- **Research Type:** Design and Development Research
- **System:** LLM-Powered Agentic Dashboard
- **Dataset:** Structured Sales Transaction Dataset
- **Architecture:** LangGraph + Dash + Streamlit
- **Evaluation:** Functional, Performance, and End-to-End Validation

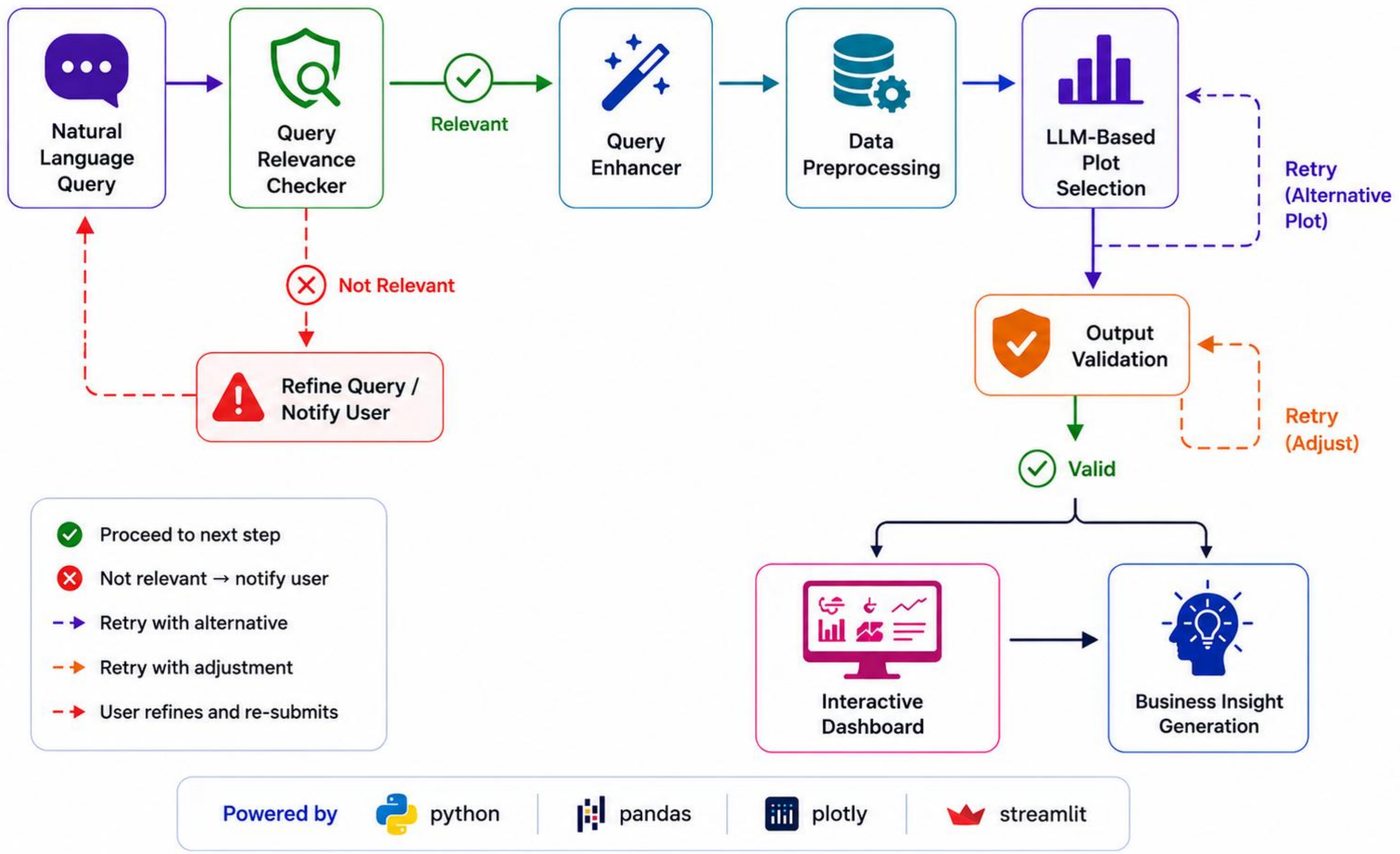
Resource Specifications

Software	Hardware	Other Resources
Python 3.9+	Intel Core i5/i7	Sales Transaction Dataset (CSV)
Pandas & NumPy	8+ GB RAM	OpenAI-Compatible API
Plotly & Dash	256 GB SSD	Stable Internet Connection
Streamlit	Windows 10/11	Environment Variables (.env)
LangGraph		

Key Highlights

- Open-source Python ecosystem
- Standard laptop hardware (No GPU required)
- Modular and scalable architecture
- Cloud-based LLM integration

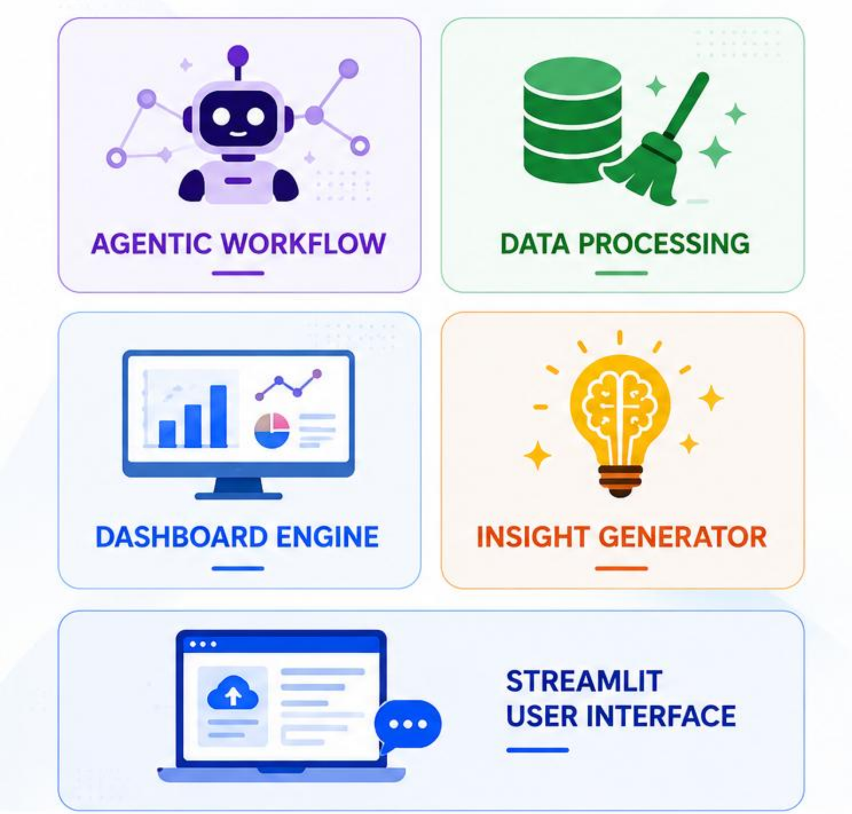
Software Design



Design Highlights

- Modular multi-agent architecture
- Validation with automatic retry logic
- Dynamic dashboard generation
- Interactive visualizations with AI-generated insights

Implementation



Implementation Deliverables

Agentic Workflow
Developed a LangGraph-based multi-agent workflow for intelligent query processing.

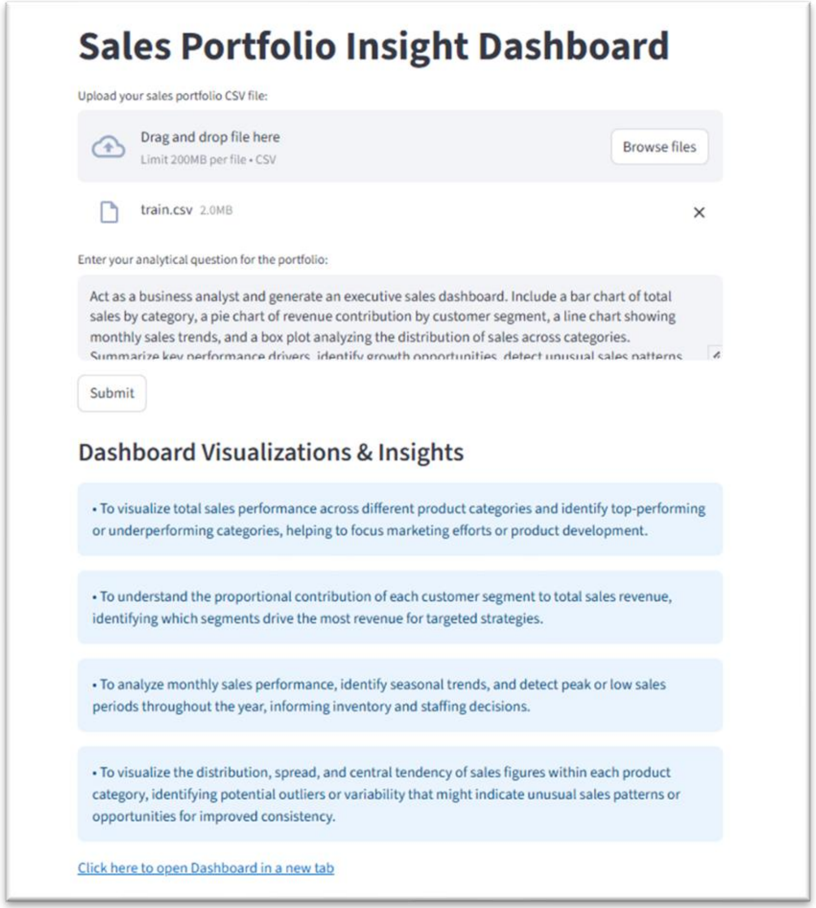
Data Processing
Implemented automated data cleaning, transformation, and preprocessing.

Dashboard Engine
Built a configuration-driven engine for dynamic dashboard generation.

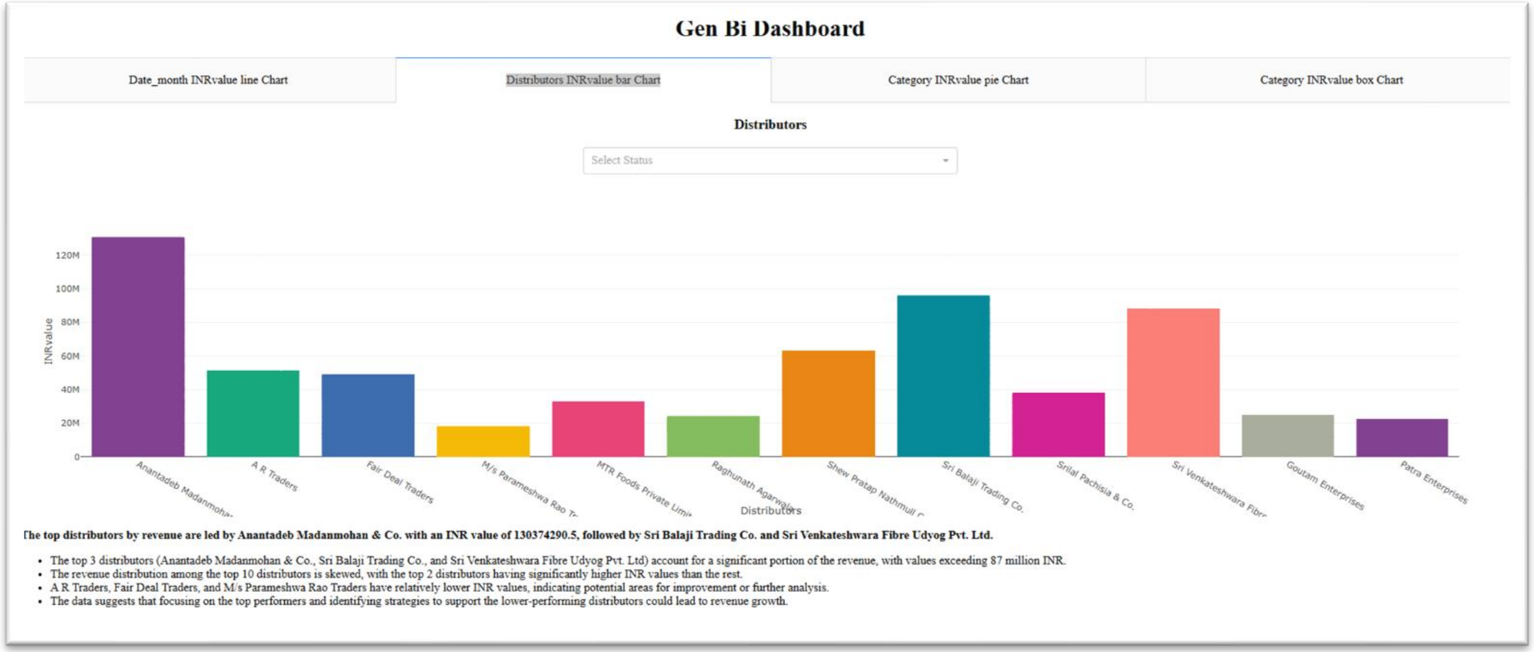
Insight Generation
Integrated LLM-powered business insight generation for visualizations.

User Interface
Developed a Streamlit interface for dataset upload and natural language interaction.

Demo



Dataset Upload & Query Input Interface



Interactive Dashboard with AI-Generated Visualizations and Business Insights

Demo

Testing Validation, and Results

Testing & Validation

Test Scenario	Expected Outcome	Status
Dataset Upload	Dataset loaded successfully	✅ Pass
Natural Language Query	Query processed correctly	✅ Pass
Dashboard Generation	Appropriate visualizations generated	✅ Pass
Interactive Filtering	Dashboard updates dynamically	✅ Pass
Insight Generation	Relevant business insights produced	✅ Pass

Key Results

- Successfully generated interactive dashboards from natural language queries.
- Automated visualization selection based on dataset characteristics.
- AI-generated contextual business insights for decision support.
- Validation and retry mechanism improved output reliability.
- Most dashboard responses were generated within **one minute**.

Conclusion and Future Scope

Conclusion

- Demonstrated the feasibility of integrating **LLMs with Business Intelligence** for automated analytics.
- Simplified the analytics workflow by reducing manual dashboard configuration.
- Enabled intuitive data exploration through **natural language interaction**.
- Established a scalable foundation for intelligent, self-service business intelligence.

Future Scope

- Integrate real-time and database-driven data sources.
- Expand support for advanced visualizations and analytical techniques.
- Enable multi-user collaboration with role-based access control.
- Support multiple LLM providers and enterprise deployment.

References

Bibliography | Webliography

1) A. Wu *et al.*, “AI4VIS: Survey on Artificial Intelligence Approaches for Data Visualization.” [Online]. Available: http://csrankings.org/

2) Y. Luo, N. Tang, G. Li, J. Tang, C. Chai, and X. Qin, “Natural Language to Visualization by Neural Machine Translation.” [Online]. Available: https://github.com/Thanksyy/ncNet

3) M. Mersha, K. Lam, J. Wood, A. AlShami, and J. Kalita, “Explainable Artificial Intelligence: A Survey of Needs, Techniques, Applications, and Future Direction,” Jan. 2025, doi: 10.1016/j.neucom.2024.128111.

4) N. Bikakis, “Big Data Visualization Tools,” in *Encyclopedia of Big Data Technologies*, Springer International Publishing, 2022, pp. 1–8. doi: 10.1007/978-3-319-63962-8_109-2.

5) A. Wu *et al.*, “MultiVision: Designing Analytical Dashboards with Deep Learning Based Recommendation,” Jul. 2021, [Online]. Available: http://arxiv.org/abs/2107.07823

6) C. North, “Toward measuring visualization insight,” *IEEE Comput. Graph. Appl.*, vol. 26, no. 3, pp. 6–9, May 2006, doi: 10.1109/MCG.2006.70.

References

Bibliography | Webliography

7) LangChain, "LangGraph: Build Stateful Multi-Actor Applications with LLMs," https://langchain-ai.github.io/langgraph/.

8) G. Ouyang *et al.*, "NVAGENT: Automated Data Visualization from Natural Language via Collaborative Agent Workflow," 2025. [Online]. Available: https://github.com/

9) S. Few, "Dashboard Design for at-a-glance monitoring," 2010. [Online]. Available: www.perceptualedge.com

10) B. Shneiderman, "The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations."

11) "Team Data Science Process," https://www.datascience-pm.com/tdsp/.

12) V. J. Hodge and J. Austin, "A Survey of Outlier Detection Methodologies," 2004.

Annexure

Additional Information | Plagiarism score

GitHub Link

GitHub Link

https://github.com/Shyamsundarkanaka/LLM-Powered-Agentic-Dashboard-for-Business-Intelligence

Thank you!